

Technical description

Grease sampling at ENERCON wind energy converters



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000	2018-08-15	-	Document created
001	2021-09-10	4.1	Results adapted
002	2021-09-29	4.1	Results adapted
003	2024-01-18	1	Analysis set defined
		3	WEC types added Taking of grease samples updated
		4	Limit values updated

General information

This document describes the grease sampling method at ENERCON wind energy converters.

The sampling method described here ensures that the grease is actually taken from the bearing, and thus that the sample is representative.

If the grease sample is taken incorrectly, this can have unnecessary and cost-intensive consequences.

The figures are intended for illustration purposes only and may differ from the actual components.

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1 Required tools/materials

Table 1: Required tools/materials

Quantity	Required tools
1	Combination spanner, A/F 14
1	Combination spanner, A/F 17
1	Ratchet ½"
1	Socket ½" hexagon socket 17 mm
Quantity	Required material
	Cleaning cloths (lint free)
	Cable tie, approx. 5 mm x 200 mm (w x l)
	Disposable plastic spatula
	Analysis kit for grease (OelCheck analysis kit 4, 5 or Wind/turquoise)

2 Safety instructions

DANGER



Operating a wind energy converter involves certain risks.

A lack of knowledge of the principle of operation of the wind energy converter and the associated risks can result in death, serious injury and damage to property.

- ⇒ Read the operating instructions of the wind energy converter carefully.
- ⇒ Observe all notes in the operating instructions of the wind energy converter, especially the safety instructions.
- ⇒ Refrain from any activities in and at the wind energy converter whose effects you cannot anticipate or which are prohibited by the operating instructions.
- ⇒ Observe the additional safety provisions applicable at the deployment site.

3 Taking grease samples

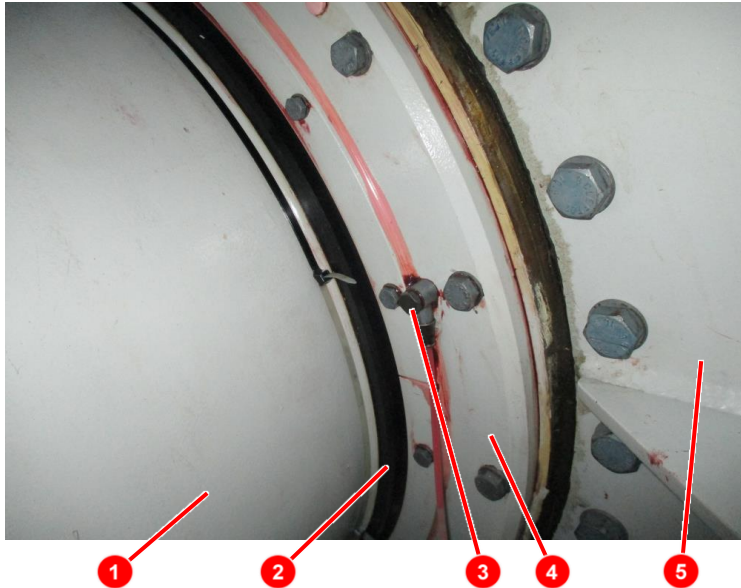


Fig. 1: Example: Rear rotor bearing E-70 E4

- 1 Axle pin
- 2 V ring
- 3 Connection of central lubrication system, A/F 17 or grease nipple, A/F 14 or lubricator
- 4 Bearing cover
- 5 Rotor

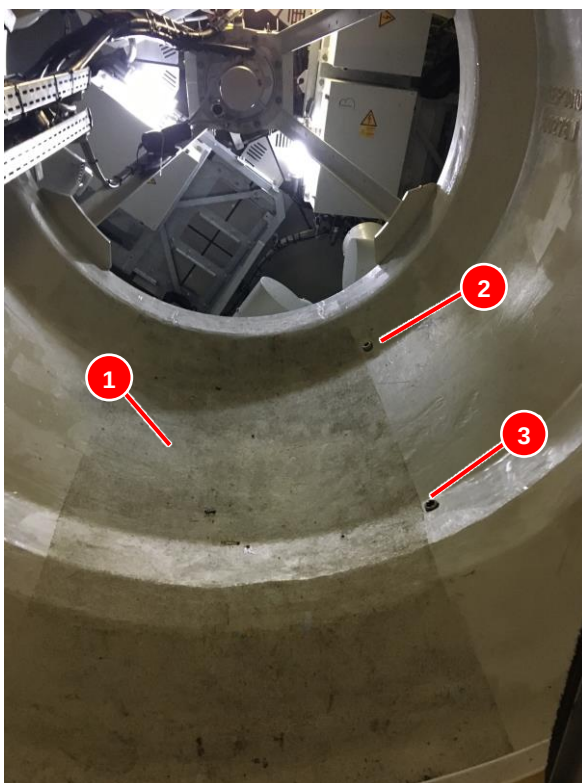


Fig. 2: Rotor bearing unit EP3

- 1 Axle pin
- 2 Sampling point of front rotor bearing
- 3 Sampling point of rear rotor bearing

NOTICE



The use of grease which has leaked from the bearing (e.g. in the area of the V ring, Fig. 1, item 2) for the grease sample is prohibited.

Any grease sampled from there is not representative for the grease in the area of the bearing raceways.

⇒ Never use grease which has leaked from the bearing for a grease analysis. This grease can be contaminated by its environment.



ENERCON accepts only grease samples which can be proven to have been taken using the correct method.

1. Thoroughly clean the area around the sampling point with lint free cloths.

2. **EP1, EP2, EP4, EP8, E-101, E-115:**

Remove one grease nipple.

or

Unscrew one lubricator.

or

Unscrew one connection of the central lubrication system.

EP3:

Unscrew an M20 cylinder head bolt in the axle pin (see Fig. 2).

3. Using a suitable aid (e.g. cable tie), take the quantity of grease required for analysis (approx. 30 g) out of the bearing. Discard fresh, unused grease stemming from the front area of the sampling point. Use several sampling points if required.

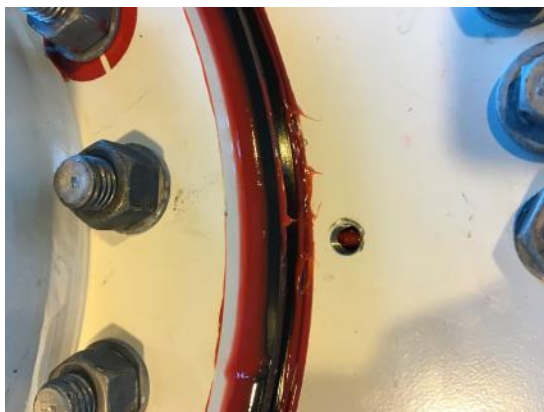


Fig. 3: Lubrication point as sampling point



Fig. 4: Grease sampling using cable tie

4. Fill the grease sample into an appropriate sample vessel.



When filling in the sampling data sheet, please note:

1. State the correct bearing:

EP2, EP4, EP8, E-101, E-115:

- The **front rotor bearing** is the tapered roller bearing at the end of the axle pin. This is also where the slip ring unit is located. **Bearing A** is an alternative name.
- The **rear rotor bearing** is a cylindrical roller bearing. It is located in the area of the generator. **Bearing B** is an alternative name.

EP1, EP3:

- The **front rotor bearing** is a tapered roller bearing at the end of the axle pin. This is also where the slip ring unit is located. **Bearing A** is an alternative name.
- **The rear main bearing**, also a tapered roller bearing, is located in the area of the generator. **Bearing B** is an alternative name.

2. State the correct grease type:

- Red grease: Mobil SHC 460 WT
- Beige grease: Mobil SHC 461 WT
- Yellow grease: Klüberplex BEM 41-141

4 Limit values for the assessment of grease samples

Based on the experience with rolling-element bearings, the following limit values may be used for the assessment according to ENERCON:

Limit values	Result
Fe < 3000 ppm PQ index < 300	There is no need for technical measures. The grease fulfils the purpose intended for use. There are no restrictions on the usability of the grease.
Fe > 3000 ppm and/or PQ index > 300	There are indications that grease is contaminated. ENERCON performs the following activities: • Check of the bearing in question • Verification of the grease sample • Flushing the bearing for decontamination of the grease



ENERCON will continually check the limit values indicated and adapt them to new findings as required.